

Unit A3: Databases (HL Extension) - Expanded Guide

Sub-topics: A3.3.4 - A3.3.6, A3.4

1. Syllabus Scope & Learning Outcomes

These topics are for Higher Level students only. The content here is more theoretical and abstract.

Syllabus Point	Command Term	Student Expectation (The "Do")
A3.3.4	Construct	Construct SQL queries using Aggregate Functions (SUM, AVG, COUNT, MAX, MIN) and grouping (GROUP BY, HAVING).
A3.3.5	Describe	Describe the utility of Views : Restricting access (Security), simplifying complex joins (Usability), and presenting computed data.
A3.3.6	Describe	Describe Transactions and the ACID properties required for data integrity. Explain COMMIT and ROLLBACK.
A3.4.1	Outline	Outline alternative models: NoSQL (Document, Key-Value), Spatial (GIS), Object-Oriented , In-Memory .
A3.4.2	Explain	Explain Data Warehousing : The separation of Operational

		Data (OLTP) from Analytical Data (OLAP).
A3.4.3	Explain	Explain Data Mining : Discovering hidden patterns using Classification, Clustering, Regression, and Association .
A3.4.4	Describe	Describe Distributed Databases: Partitioning (Horizontal/Vertical), Replication (Master-Slave), and Transparency .

2. Key Terminology & Definitions

- **Aggregate Function:** A function that performs a calculation on a set of values to return a single value.
- **GROUP BY:** An SQL clause used with aggregates to group rows that have the same values into summary rows.
- **HAVING:** Like a WHERE clause, but for groups (e.g., show groups with COUNT > 5).
- **View:** A stored query that looks like a table. It provides a specific "view" of the data without duplicating it.
- **Transaction:** A unit of work (sequence of operations) performed against a database.
- **ACID Properties:**
 - **Atomicity:** The "All or Nothing" rule. If one part fails, the whole transaction fails.
 - **Consistency:** The transaction takes the database from one valid state to another valid state.
 - **Isolation:** Concurrent transactions do not affect each other.
 - **Durability:** Once committed, data survives system crashes.
- **Data Warehouse:** A centralized repository for accumulated historical data from various sources, optimized for reporting.
- **ETL (Extract, Transform, Load):** The process of moving data from operational systems to a warehouse.
- **Data Mining:** The automated process of discovering patterns/knowledge from large datasets.
 - **Association Rule:** Finding relationships between variables (e.g., Market Basket Analysis).
 - **Cluster Analysis:** Grouping a set of objects in such a way that objects in the same

group are more similar to each other.

- **Distributed Database:** Portions of the database are stored in multiple physical locations.

3. Core Content & Exam Actions

A3.3.4 - A3.3.6: Advanced SQL & Integrity

Theory:

- **Aggregates:** Essential for reporting. `SELECT Dept, AVG(Salary) FROM Employees GROUP BY Dept;`
- **Views:** Used to hide sensitive columns (e.g., hiding SocialSecurityNumber from a receptionist view).
- **ACID Compliance:**
 - Example: Transferring money.
 - Step 1: Deduct \$100 from A.
 - Step 2: Add \$100 to B.
 - **Atomicity** ensures that if Step 2 fails, Step 1 is rolled back.

Student Exam Action:

- **Coding:** Write a query to find departments with more than 10 employees.
`SELECT Department, COUNT(EmployeeID)
FROM Staff
GROUP BY Department
HAVING COUNT(EmployeeID) > 10;`
- **Explanation:** Explain why **Durability** is critical for a bank (records must persist even if power is lost immediately after a transaction confirms).

A3.4: Advanced Database Models

Theory:

- **Data Warehousing:**
 - **Subject-Oriented:** Focused on sales, marketing, etc., not processes.
 - **Integrated:** Data from different sources is made consistent.
 - **Time-Variant:** Historical data is kept (unlike operational DBs which overwrite).
 - **Non-Volatile:** Data is loaded and accessed, not updated.
- **Data Mining Techniques:**
 - **Classification:** Predicting a category (e.g., "Spam" or "Not Spam").
 - **Clustering:** Finding natural groups (e.g., Customer Segmentation).
 - **Regression:** Predicting a number (e.g., House Price prediction).
 - **Association:** "If A, then B" (e.g., If Bread, then Butter).
- **Distributed DBs:**
 - **Horizontal Partitioning (Sharding):** Rows 1-1000 in London, 1001-2000 in NY.
 - **Vertical Partitioning:** Columns Name, Address in Server A, Salary, History in Server

B.

- **Replication:** Copying the whole DB to multiple servers for redundancy and faster reads.

Student Exam Action:

- **Comparison:** Compare **OLTP** (Online Transaction Processing - Day to Day operations) with **OLAP** (Online Analytical Processing - Warehousing/Analysis).
- **Evaluation:** Evaluate the trade-off of **Replication** in a distributed database (Higher availability/read speed vs. Slower write speed/data inconsistency risk).

4. Common Pitfalls & Examiner Tips

- **Pitfall (HAVING vs WHERE):** WHERE filters rows *before* grouping. HAVING filters groups *after* grouping.
- **Pitfall (Data Mining):** Confusing **Clustering** (grouping without predefined categories) with **Classification** (assigning to predefined categories).
- **Tip (Warehouse):** Remember the keyword **Time-Variant**. Operational DBs show "Now". Warehouses show "History".